

Question

The polymer prepared by a certain method exhibits a mole fraction and degree of polymerization that follow the following Poisson distribution.

$$x_k = \frac{\lambda^k}{k!} e^{-\lambda}$$

Where x_k is the ratio of the amount of substance of polymer chains with a degree of polymerization of k to the total amount of substance of polymer chains, k is the degree of polymerization, and λ is the reaction ratio of monomer to initiator. Assuming that the relative molecular mass of the polymer is $M_k = M_0 + kM$. Then, for the following sets of parameters, the polymer molecular weight distribution that is the most uniform is:

A. All other options are incorrect

B. $\lambda = 25, M_0 = 100, M = 80$

C. $\lambda = 25, M_0 = 100, M = 100$

D. $\lambda = 25, M_0 = 200, M = 55$

E. $\lambda = 25, M_0 = 200, M = 75$

F. $\lambda = 25, M_0 = 400, M = 60$

G. $\lambda = 25, M_0 = 400, M = 76$

H. $\lambda = 50, M_0 = 40, M = 55$

I. $\lambda = 50, M_0 = 40, M = 65$

J. $\lambda = 50, M_0 = 70, M = 50$

K. $\lambda = 50, M_0 = 70, M = 60$

L. $\lambda = 50, M_0 = 90, M = 48$

M. $\lambda = 50, M_0 = 90, M = 57$

Answer

Correct Answer: **L**

Detailed Explanation

The physical quantity that can reflect the molecular weight distribution of a polymer is the ratio of weight-average molecular weight to number-average molecular weight, $\frac{X_w}{X_n}$. Therefore, it is necessary to first derive the expression for $\frac{X_w}{X_n}$.

CHECKPOINT

1 PTS

$\frac{X_w}{X_n}$ can reflect the molecular weight distribution of the polymer

According to the definition, we have $\frac{X_w}{X_n} = \frac{\sum n_k \sum n_k M_k^2}{(\sum n_k M_k)^2} = \frac{\sum x_k M_k^2}{(\sum x_k M_k)^2}$

CHECKPOINT

1 PTS

$$\frac{X_w}{X_n} = \frac{\sum x_k M_k^2}{(\sum x_k M_k)^2}$$

Where $\sum x_k M_k = M_0 \sum \frac{\lambda^k}{k!} e^{-\lambda} + M \sum k \frac{\lambda^k}{k!} e^{-\lambda}$

And we have $M_0 \sum \frac{\lambda^k}{k!} e^{-\lambda} = M_0$

$M \sum k \frac{\lambda^k}{k!} e^{-\lambda} = M \lambda e^{-\lambda} \sum \frac{\lambda^{k-1}}{(k-1)!} = M \lambda$

Therefore, we have $\sum x_k M_k = M_0 + M \lambda$

CHECKPOINT

1 PTS

$$\sum x_k M_k = M_0 + M \lambda$$

In addition, we have

$$\sum x_k M_k^2 = M_0^2 \sum \frac{\lambda^k}{k!} e^{-\lambda} + 2 M M_0 \sum k \frac{\lambda^k}{k!} e^{-\lambda} + M^2 \sum k^2 \frac{\lambda^k}{k!} e^{-\lambda}$$

Where

$$M_0^2 \sum \frac{\lambda^k}{k!} e^{-\lambda} = M_0^2$$

$$2 M M_0 \sum k \frac{\lambda^k}{k!} e^{-\lambda} = 2 M M_0 \lambda$$

Specifically, we have

$$\sum k^2 \frac{\lambda^k}{k!} = \sum k \frac{\lambda^k}{(k-1)!} = \sum \frac{k}{k-1} \times \frac{\lambda^k}{(k-2)!} = \lambda^2 \sum \frac{\lambda^{k-2}}{(k-2)!} + \lambda \sum \frac{\lambda^{k-1}}{(k-1)!} = e^\lambda (\lambda^2 + \lambda)$$

From this, we have

$$M^2 \sum k^2 \frac{\lambda^k}{k!} e^{-\lambda} = M^2 (\lambda^2 + \lambda)$$

CHECKPOINT

0.5 PTS

$$M^2 \sum k^2 \frac{\lambda^k}{k!} e^{-\lambda} = M^2 (\lambda^2 + \lambda)$$

Therefore, we have

$$\sum x_k M_k^2 = M_0^2 + 2MM_0\lambda + M^2(\lambda^2 + \lambda)$$

CHECKPOINT

1 PTS

$$\sum x_k M_k^2 = M_0^2 + 2MM_0\lambda + M^2(\lambda^2 + \lambda)$$

Therefore, we obtain

$$\frac{X_w}{X_n} = 1 + \frac{\lambda M^2}{(M_0 + M\lambda)^2}$$

CHECKPOINT

2 PTS

$$\frac{X_w}{X_n} = 1 + \frac{\lambda M^2}{(M_0 + M\lambda)^2}$$

The value of $\frac{X_w}{X_n}$ can reflect the molecular weight distribution of the polymer; the smaller the value, the more uniform the distribution.

CHECKPOINT

1 PTS

The smaller the $\frac{X_w}{X_n}$, the more uniform the molecular weight distribution of the polymer

Substituting the data in the question into the calculation, the values of $\frac{X_w}{X_n} - 1$ corresponding to each option are $B = 0.03628$; $C = 0.03698$; $D = 0.03049$; $E = 0.03266$; $F = 0.02493$; $G = 0.02730$; $H = 0.01943$; $I = 0.01952$; $J = 0.01893$; $K = 0.01910$; $L = 0.01858$; $M = 0$.

The value of option L is the smallest, which is 0.01858